

CHAPTER 6

Understanding the Response Structure and Process in the Polytomous Rasch Model

David Andrich

The University of Western Australia

Editor Introduction: Rather than developing a specific polytomous IRT model, this chapter outlines and argues for the importance of the item response process that is modeled by all polytomous IRT models in the Rasch family of models. Modeling response processes in this way is argued to have an important advantage over the way the response process is modeled in what Hambleton, van der Linden, and Wells (Chapter 2) call indirect models and what are elsewhere referred to as divide-by-total (Thissen & Steinberg, 1986) or adjacent category (Mellenberg, 1995) models.

The Rasch model for ordered response categories in standard formats was derived from a sequence of theoretical propositions requiring invariance of comparisons among item and among person parameter estimates. A model with sufficient statistics is the consequence. The model was not derived to describe any particular data (Andersen, 1977; Andrich, 1978; Rasch, 1961).

Standard formats involve only one response in one of the categories deemed a priori to reflect *increasing* levels of the property and are common in quantification of performance, status, and attitude in the social sciences. The advantage of an item with more than two ordered categories is that, if the categories work as intended, it gives more information than a dichotomous item. Table 6.1 shows three common examples. Figure 6.1 shows a graphical counterpart using the first example in Table 6.1.

The ordered categories on the hypothesized continuum in Figure 6.1 are contiguous. They are separated on the continuum by successive points termed *thresholds*. This is analogous to mapping a location of an object on a line partitioned into equal units to obtain a physical measurement. Because we do not have a fixed origin, there is no endpoint on the latent continuum of

TABLE 6.1 Standard Response Formats for the Rasch Model

Fail	Pass	Credit	Distinction
Never	Sometimes	Often	Always
Strongly disagree	Disagree	Agree	Strongly agree

Figure 6.1 for the extreme categories, only partitions of the continuum into four contiguous categories, which requires three thresholds.

Terminology Note: Thresholds are also sometimes referred to as category boundaries. What this chapter makes clear, and what should always be remembered, is that irrespective of which term is used (*category boundary* or *threshold*), these points on the trait continuum, which separate ordered response categories, are defined very differently in cumulative (e.g., Samejima logistic graded response model) or adjacent category (e.g., Rasch) models. As a result, these category-separating points have a different meaning in these two most common types of polytomous IRT models.

In elementary analyses of data obtained from formats such as those in Table 6.1, and by analogy to physical measurement, successive integers are assigned to the categories. In more advanced analyses, a probabilistic model that accounts for the finite number of categories and for sizes of the categories is applied. The Rasch model is one such model, and in this model the successive categories are scored with successive integers.

Identity of Rating and Partial Credit Models

In the examples in Table 6.1 it might be considered that if the same format is used across all items, the sizes of the categories will also be the same across all items. However, that is an empirical question, and it is possible that there is an interaction between the content of the item and the response format, so that the sizes of the categories are different for different items. It may also be the case that different items have different formats that are natural to the item with different numbers of categories, as, for example, when there are different items in an achievement test with different maximum scores. The Rasch model that has the same format across all items *and* has the same sized

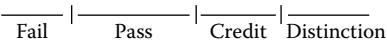


FIGURE 6.1 Graphical representation of ordered categories.

categories is referred to sometimes as the *rating scale model*. The model with different sized categories or with different numbers of categories is referred to sometimes as the *partial credit model*. The difference, as will be seen, is only a matter of parameterization, and at the level of a single person responding to a single item, the models are identical. The focus of this chapter is on the response of one person to one item that covers both parameterizations. Therefore, the model will be referred to simply as the polytomous Rasch model (PRM). The dichotomous model is simply a special case of the PRM, but where it is necessary to distinguish it as a dichotomous model in the exposition, it will be referred to explicitly as the *dichotomous RM*.

Distinctive Properties of the PRM

The PRM has two properties that, when first disclosed, were considered somewhat counterintuitive: First, combining adjacent categories by summing the probabilities of responses in the categories, and in the related sense of summing their frequencies to form a single category, can only be done under very restricted circumstances (Andersen, 1977; Andrich, 1978, Jansen & Roskam, 1986; Rasch, 1966). Second, the thresholds analogous to those in Figure 6.1 that define the boundaries of the successive categories may take on *values* that are not in their natural order. In part because these properties are exactly opposite to those of the then prevailing model for ordered categories, that based on the work of Thurstone (Thurstone & Chave, 1929), they have been ignored, denied, circumvented, or generated debate and misunderstandings in the literature (Andrich, 2002). Known in psychometrics as the *graded response model* (GRM), the latter model has been developed further by Bock (1975), Samejima (1969, 1997, 1996), and McCullagh (1980).

Criterion That Ordered Categories Should Satisfy

One observation from these reactions to the model's properties is that in the development of response models for ordered category formats there is no *a priori* articulation of any criterion that data in ordered categories should satisfy—it seems it is simply assumed that if categories are deemed to be ordered, they will necessarily operate that way. One factor that immediately comes to mind as possibly violating the required order is respondents not being able to distinguish between two adjacent categories. This has been observed in data in Andrich (1979).

The theme of this chapter is that it is an empirical hypothesis whether or not ordered categories work as intended. The chapter sets up a criterion that must be met by data in an item response theory framework for it to be evident empirically that the categories are working as intended, and shows how the PRM makes a unique contribution to providing the empirical evidence. Meeting this requirement empirically is necessary because if the intended ordering of the categories does not reflect successively more of the property,

then it puts into question the very understanding of what it means to have more of the property and of any subsequent interpretations from the data. The chapter is not concerned with issues of estimation and the tests of fit, which are well covered in the literature, but in better understanding the distinctive properties of the model itself, and the opportunities it provides for the empirical study of ordered polytomous response formats.

It is stressed that the criterion for ordered categories working as intended *pertains to the data*, and not to response models themselves *irrespective of the data*. The importance of distinguishing between the properties of data from procedures of models of analysis for ordered categories was recognized by R. A. Fisher, who had a method for analyzing data intended to be in ordered categories, and upon obtaining results for a particular data set noted: “It will be observed that the numerical values ... lie ... in the proper order for increasing reaction. This is not a consequence of the procedure by which they have been obtained, but a property of the data examined” (Fisher, 1958, p. 294). Any note from Fisher is worthy of substantial consideration and study (Wright, 1980).

This chapter demonstrates that the properties of the PRM are compatible with treating the operation of the ordered categories as an empirical hypothesis. In particular, it is demonstrated that the model has the remarkable property that from a set of structurally dependent responses in an ordered category format, *it recovers information that would arise from compatible, experimentally independent formats*. This permits the inference regarding the empirical ordering of categories. Thus, the chapter does not merely describe the Rasch model for ordered categories from the perspective of modeling data and for providing invariant comparisons, but presents a case that it is the ideal model for characterizing the intended response process and for testing empirically whether ordered categories are operating as required.

The chapter is organized as follows. We first describe an experiment with independent responses at thresholds devised to assess unequivocally the empirical ordering of the categories. We then analyze in detail three response spaces, whose relationship needs to be understood. We also explain why the probabilities and frequencies in adjacent categories cannot be summed in the PRM except in special circumstances. Finally, we conclude with a summary that includes a suggestion as to why over the long history of the development and application of models for data in ordered categories, and despite the lead from Fisher, no previous criteria have been articulated that ordered categories must meet.

Criteria for Data in Ordered Response Categories

In preparation for developing and specifying a criterion for the empirical ordering of categories, we consider some relationships between models and data. These relationships are generally taken for granted, but they are made explicit here because of their specific roles in relation to the PRM and the theme of the chapter.

A Comment on the Uses of Models

One use of models is simply to summarize and describe data. Models *describe* data in terms of a number of parameters that are generally substantially smaller than the number of data points. It is of course necessary to check the fit between the data and the model to be satisfied that the model does describe the data.

A second use of models is to characterize the *process* by which data are generated. For example, the Poisson distribution arises from, among many other circumstances, the “cumulative effect of many improbable events” (Feller, 1950, p. 282). This model is derived a priori to the data in characterizing a response process. If the data do not fit the model, then a question might be asked about its characterization of the process. However, the fit of the data to the model is only a necessary, not sufficient, condition to confirm that the model characterizes the process in those data.

A third and much less conventional use of models is to express a priori conditions that data are required to follow if they are to subscribe to some principles. As indicated above, this is the case with the PRM. Following a series of studies, Rasch articulated conditions of invariance of comparisons that data should have if they are to be useful in making quantitative statements. Specifically, (1) the comparison between two stimuli should be independent of which particular individuals were instrumental for the comparison; (2) symmetrically, a comparison between two individuals should be independent of which particular stimuli within the class considered were instrumental for comparison (Rasch, 1961, p. 332).

These conditions of invariance were not unique to Rasch—virtually identical conditions were articulated by Thurstone (1928) and Guttman (1950) before him. However, the distinctive contribution of Rasch was that Rasch expressed these conditions in terms of a probabilistic model. Rasch wrote his conditions for invariance in a general equation, which, in the probabilistic case for dichotomous responses, takes the form

$$P\{(Y_{ni} = y_{ni}, Y_{nj} = y_{nj}); \beta_n, \delta_i, \delta_j | f(y_{ni}, y_{nj})\} = \vartheta(y_{ni}, y_{nj}, \delta_i, \delta_j) \quad (6.1)$$

where Y_{ni}, Y_{nj} are random variables whose responses (y_{ni}, y_{nj}) take the values $\{0,1\}$, β_n and δ_i, δ_j are location parameters of person n and items i and j respectively, and the right side of Equation 6.1 is independent of the person parameter β_n . As indicated already, this leads to a class of models with sufficient statistics for the parameters, which generalizes to the PRM.

The key advantage of specifying the conditions in terms of a model is that mathematical consequences, some of which might be initially counterintuitive, can be derived. This is the case with the Rasch model. However, when the consequences follow mathematically from a specification as compelling as that of making relatively invariant comparisons, then because they can provide genuinely new insights that might not be apparent immediately intuitively, they should be understood. Another distinctive consequence of this

use of a model is that no amount of data analysis and demonstration of fit to the model or otherwise is relevant to the *case for the model*.

Definition of a Threshold in the Presence of Experimental Independence

Suppose that it is intended to assess the relative location of persons on some construct that can be mapped on a linear continuum, for example, an achievement test. Items of successively increasing difficulty would be “landmarks” of achievement (Thurstone, 1925) requiring successively increasing ability for success.

Suppose further that the responses of persons to items are scored dichotomously, for a *successful* and *unsuccessful* response, respectively. From such responses, and in an arbitrary unit, the dichotomous RM (Fischer & Molenaar, 1995; Rasch, 1960, 1961; Wright, 1997; Wright & Panchapakesan, 1969) can be used to estimate the relative location of items on the continuum. This model takes the form

$$P\{Y_{ni} = y\} = \frac{e^{y(\beta_n - \delta_i)}}{1 + e^{\beta_n - \delta_i}} \tag{6.2}$$

where the variables are identical to those of Equation 6.1. The response function of Equation 6.2 for $y = 1$ is known as the item characteristic curve (ICC). Three ICCs for the dichotomous RM are illustrated in Figure 6.2. The data giving rise to estimates in Figure 6.2 were simulated for 5,000 persons responding to six items independently (two sets of three items) with locations of $-1.0, 0.0$, and 1.0 , respectively, in the first set. Only the responses of the first set of three items are shown in Figure 6.1. These data, together with those of the second set, are used later in the chapter to illustrate the derivations.

The responses $\{0,1\}$ are ordered; response $y = 1$ is deemed successful, and the response $y = 0$ unsuccessful. In achievement testing δ_i is referred to as

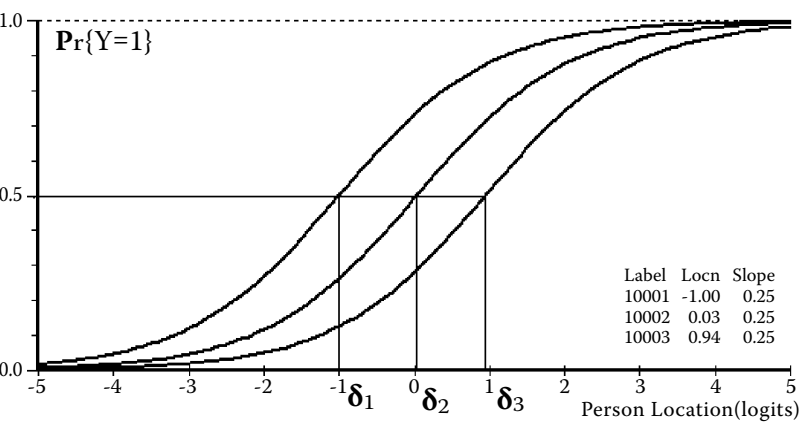


FIGURE 6.2 ICCs for three items.

Copyright © 2010. Taylor & Francis Group. All rights reserved.

the difficulty of the item. In general terms, and following psychophysics (Bock & Jones, 1968), it is termed a *threshold*—it is the point at which the person with the same location $\beta_n = \delta_i$ has an equal probability of being successful and unsuccessful:

$$\Pr\{Y_{ni} = 1\} = \Pr\{Y_{ni} = 0\} = 0.5 \tag{6.3}$$

In the dichotomous RM, the ICCs are parallel (Wright, 1997), which is exploited in this chapter. We use the dichotomous RM to construct the PRM. However, to better understand the PRM for more than two ordered categories, the two-parameter logistic model (2PLM),

$$P\{Y_{ni} = y\} = \frac{e^{y\alpha_i(\beta_n - \delta_i)}}{1 + e^{\alpha_i(\beta_n - \delta_i)}} \tag{6.4}$$

where α_i , known as the discrimination parameter, characterizes the slope of the ICC (Birnbaum, 1968), is also used in a later section of this chapter.

Notational Difference: The parameters in Equation 6.4 are presented in a format consistent with the rest of the chapter. Typically, however, the letter a is used to denote the discrimination parameter, which is denoted by α in Equation 6.4. Similarly, the 2PLM typically uses θ in place of β , and b is used for the item difficulty parameter, which is δ in Equation 6.4.

The Guttman Structure and the Dichotomous Rasch Model

Another, more explicit view of the location of items of increasing difficulty, equivalent to Thurstone’s notion of landmarks, is that of Guttman (1950), who enunciated an idealized *deterministic* response structure for unidimensional items. The Guttman structure is central in understanding the PRM.

For I dichotomous items responded to independently, there are 2^I possible response patterns. These are shown in Table 6.2 for the case of three items. The top part of Table 6.2 shows the subset of *patterns* of responses according to the Guttman structure. The number of these patterns is $I + 1$.

The rationale for the Guttman structure in Table 6.2 (Guttman, 1950) is that for unidimensional responses across items, if a person succeeds on an item, then the person should succeed on all items that are easier than that item, and that if a person fails on an item, then the person should fail on all items more difficult than that item. The content of the items with different difficulties operationalizes the continuum.

With experimentally independent items, it is possible that a deterministic Guttman structure will not be observed in data. In that case, the dichotomous RM may be used to locate items on a continuum. The dichotomous RM is a probabilistic counterpart of the Guttman structure that is a deterministic limiting case (Andrich, 1985). Specifically, for any person, the probability of

TABLE 6.2 The Guttman Structure With $I = 3$ Dichotomous Items in Threshold Order

Items	1	2	3	Total Score x	$\Pr\{y_{n1}, y_{n2}, y_{n3} x\}$
$I + 1 = 4$ Guttman Response Patterns					
	0	0	0	0	1
	1	0	0	1	0.667
	1	1	0	2	0.678
	1	1	1	3	1
$2^I - I - 1 = 4$ Non-Guttman Response Patterns					
	0	1	0	1	0.248
	0	0	1	1	0.085
	1	0	1	2	0.235
	0	1	1	2	0.087

success on an easier item will always be greater than the probability of success on a more difficult item. This statement is evident from the parallel ICC curves in Figure 6.2.

In the Guttman structure, as is evident in Table 6.2, the total score, $x, x = \sum_{i=1}^I y_i$, completely characterizes the response pattern. In the dichotomous RM, the total score plays a similar role, though probabilistically; it is a sufficient statistic for the person parameter (Andersen, 1977; Rasch, 1961). If item thresholds are then ordered in difficulty, and for a given total score, the Guttman pattern has the greatest probability of occurring (Andrich, 1985).

Furthermore, because of sufficiency, the probability of any pattern, given the total score x , is independent of the person's ability. Thus, the probabilities of the patterns of responses for total scores of 1 and 2, shown in Table 6.2, are given by

$$P\{(y_{n1}, y_{n2}, y_{n3}) | x = 1\} = \frac{e^{-y_{n1}\delta_1 - y_{n2}\delta_2 - y_{n3}\delta_3}}{e^{-\delta_1} + e^{-\delta_2} + e^{-\delta_3}} \tag{6.5}$$

$$P\{(y_{n1}, y_{n2}, y_{n3}) | x = 2\} = \frac{e^{-y_{n1}\delta_1 - y_{n2}\delta_2 - y_{n3}\delta_3}}{e^{-\delta_1 - \delta_2} + e^{-\delta_2 - \delta_3} + e^{-\delta_1 - \delta_3}} \tag{6.6}$$

respectively, both of which are independent of the person ability β_n and are special cases of Equation 6.1. These equations are the basis of conditional estimation of the item parameters independently of the person parameters (Andersen, 1973).

Design of an Experiment to Assess the Empirical Ordering of Categories

We now consider the design of an experiment in which the empirical ordering of the categories can be investigated. The key feature of this experiment is the empirical *independence* among the judgments.

TABLE 6.3 Operational Definitions of Ordered Classes for Judging Essays and a Response Structure Compatible With the Rasch Model

Fail (F)	<i>Inadequate setting:</i> Insufficient or irrelevant information given for the story. Or, sufficient elements may be given, but they are simply listed from the task statement, and not linked or logically organized.
Pass (P)	<i>Discrete setting:</i> Discrete setting as an introduction, with some details that also show some linkage and organization. May have an additional element to those listed that is relevant to the story.
Credit (C)	<i>Integrated setting:</i> There is a setting that, rather than simply being at the beginning, is introduced throughout the story.
Distinction (D)	<i>Integrated and manipulated setting:</i> In addition to the setting being introduced throughout the story, pertinent information is woven or integrated so that this integration contributes to the story.

The labels of fail, pass, credit, and distinction have been added in this table for the purposes of this chapter (Harris, 1991).

To make the case concrete, consider the ordered category descriptors shown in Table 6.3 that were used in assessing the abilities of students to write a narrative in relation to a particular criterion. The responses among the categories are not independent in the sense that if a response is made in one category, it is not made in any other category. The task is to construct a design compatible with Table 6.3 in which independence of judgments prevails.

Clearly from the descriptors for each category, there is an intended ordering in the quality of performance with respect to the feature of *setting*. We take it that the category descriptors operationalize the writing variable to be measured and describe the qualities that reflect successively better writing on this continuum. We further note that the first, and least demanding, category is a complement to the second category, and that the other categories show increasing quality of writing with respect to setting. We shall see how this complementarity of the first category to the others plays out.

The experimental design involves taking the descriptors in Table 6.3 and constructing *independent* dichotomous judgments at thresholds that are of increasing difficulty.

Instead of one judge assigning an essay into one of four categories, consider a design with three judges where each judge only declares whether each essay is successful or not in achieving the standard at one of pass, credit, or distinction. Thus, we have three *independent* dichotomous random variables. Although there are four categories, there are only three independent responses. The F descriptor helps in understanding the variable in the region of fail/pass, and helps the judge decide on the success or otherwise of the essay at this standard. This is the role of the F descriptor in this design.

We now consider this experimental design, summarized in Table 6.4, more closely.

The descriptors, as already indicated, describe the variable and what it takes to reflect more of its property. The better the essay in terms of these characteristics, the greater the probability that it will be deemed successful at each level.

TABLE 6.4 Experimental Design Giving Independent Responses

	Inadequate Setting F	Discrete Setting P	Integrated Setting C	Integrated and Manipulated Setting D
Judgment 1	Not P	P		
Judgment 2		δ_P Not C	C	
Judgment 3			δ_C Not D	D
				δ_D

Two further specific cases may be highlighted in order to make clear the operation of the experimental design. First, suppose the judge considers that the essay does satisfy the P descriptor, but observes that it does not meet the C and D descriptors. Then the judge should classify the essay as a success. Second, suppose that the judge, still with respect to success at P, considers that an essay satisfies even the qualities described in C or D, or some combination of these. Because of the structure of the descriptors as ordered categories, which implies that C reflects more of the property to be measured than P, and D even more of the property than C, the judge must classify it as a success at P. The more of the properties of C and D an essay has, the greater the probability of it being classified successful at the P level. Similar interpretations follow for decisions at each of the other categories. It is stressed that in each judgment it is the latent *continuum* that has been dichotomized at each threshold, and not the categories as such.

Requirements of Data From the Experimental Design

In such an experimental design it would be *required* that the success rate at P is greater than that at C, and that the success rate at C is in turn greater than that at D. That is, it is required that it is more difficult to be successful at D than at C, which in turn is more difficult than being successful at P.

If that were not the case, for example, the success rate at C was the same as that at D for the same essays, then it would be inferred that the judges do not distinguish between the two levels consistently. This could arise, for example, if the judge at C were harsher than intended and the judge at D were more lenient than intended. Thus, it may be that the experiment did not work, and it would need to be studied further to understand why this is the case. But such evidence is central to treating the ordering of the categories as an empirical hypothesis to be tested in the data.

Not only would we require that the thresholds increase in difficulty with their a priori ordering, but we would want the same distance between them irrespective of the location of the essays on the continuum. It seems untenable that these distances are different for essays of different quality. This uniformity of the relationships between these levels in probability is guaranteed if the success rate curves at different levels of quality of the essays are parallel, that is, if the dichotomous responses at the corresponding thresholds

follow the dichotomous RM. This is the essential justification for applying the dichotomous RM to such data.

In summary, if $\delta_p, \delta_c, \delta_d$ are the difficulties of the thresholds at the criteria of pass, credit, and distinction, respectively, then we require the data to fit the dichotomous RM and that $\hat{\delta}_p < \hat{\delta}_c < \hat{\delta}_d$.

We stress that there is no artificial constraint in the design, or in the model of analysis, that would ensure the apparent correct ordering of the thresholds no matter the properties of the data. As required by Fisher, the ordering of thresholds and fit to the dichotomous RM would be a property of the data, not a constraint on the design or the procedure of analysis.

The suggested empirical design of Table 6.4 may not be constructed in any particular data collection. However, it is argued here that any design that answers whether or not the ordering of the categories is working as intended *should be equivalent to this design*.

Construction and Analyses of Outcome Spaces

We now formalize some concepts and notation concerned with response spaces. Response spaces are not a typical concern of psychometrics, but they are essential in understanding the PRM. For purposes of exposition, we proceed first from the case of experimentally independent responses at the successive *thresholds*, as in Table 6.4, to one in which the responses in categories are not independent, as in Table 6.3. We then reverse the direction of derivations from the design of Table 6.3, in which a response in *categories* is dependent, to infer experimentally independent responses at *thresholds*, as in Table 6.4.

The Experimentally Independent Outcome Space Ω

Let $\{Y_{nix} = y\}, y \in \{0, 1\}$ be m_i experimentally independent random variables of the response for person n with respect to $x = 1, 2, \dots, m_i$ thresholds for item i . For example, we assume that an item such as that in Table 6.3 has been resolved into three independent judgments at successive categories, beginning with the second category, as in the design of Table 6.4. As noted earlier, with four categories, we can have only three independent responses. Let

$$P\{Y_{nix} = 1\} \equiv P_{nix}; 1 - P_{nix} \equiv Q_{nix} \quad (6.7)$$

Let $\Omega \equiv \{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})\}$ be the outcome space for all 2^{m_i} possible response patterns across the x thresholds. For example, in the case of three thresholds, as in Tables 6.2 and 6.4, there are $2^3 = 8$ possible response patterns. These, and the corresponding probabilities, are shown in Table 6.5. With experimental independence, the probability of any pattern $(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})$ is given by

$$P\{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})\} = \prod_{x=1}^{m_i} P_{nix}^{y_{nxi}} Q_{nix}^{1-y_{nxi}} \quad (6.8)$$

In the case of the response pattern (1,1,0) for three thresholds, for example,

$$P\{(1,1,0)\} = P_{ni1}P_{ni2}Q_{ni3}$$

Clearly,

$$\sum_{\Omega} P\{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})\} = \sum_{\Omega} \prod_{x=1}^{m_i} P_{nix}^{y_{nix}} Q_{nix}^{1-y_{nix}} = 1 \quad (6.9)$$

The outcome space Ω and probabilities of all outcomes are also shown in Table 6.5.

Because it proves critical to keep track of the outcome space of a response, we formalize Equation 6.7 as

$$\Pr\{Y_{nix} = 1|\Omega\} \equiv P_{nix}; \Pr\{Y_{nix} = 0|\Omega\} = 1 - P_{nix} \equiv Q_{nix} \quad (6.10)$$

In the case where the thresholds are ordered in difficulty, let

$$(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i}) = \left(\underbrace{1, 1, 1, \dots, 1}_x, \underbrace{0, 0, 0, \dots, 0}_{m_i-x} \right)$$

TABLE 6.5 The Experimentally Independent Outcome Space Ω and the Guttman Subspace Ω'

Y_{n1i}	Y_{n2i}	Y_{n3i}	
0	0	0	
Q_{ni1}	Q_{ni2}	Q_{ni3}	
1	0	0	
P_{ni1}	Q_{ni2}	Q_{ni3}	
1	1	0	
P_{ni1}	P_{ni2}	Q_{ni3}	
1	1	1	Ω'
P_{ni1}	P_{ni2}	P_{ni3}	
0	1	0	
Q_{ni1}	P_{ni2}	Q_{ni3}	
0	0	1	
Q_{ni1}	Q_{ni2}	P_{ni3}	Ω
1	0	1	
P_{ni1}	Q_{ni2}	P_{ni3}	
0	1	1	
Q_{ni1}	P_{ni2}	P_{ni3}	

$$\sum_{\Omega} P\{(y_{n1i}, y_{n2i}, \dots, y_{nki}, \dots, y_{nim_i})\} = \sum_{\Omega} \sum_k^3 \prod_{k=1}^{m_i} P_{nki}^{y_{nki}} Q_{nki}^{1-y_{nki}} = 1$$

be a sequence of responses in which the first x are successes, followed by $m_i - x$ failures. That is, the response is a Guttman pattern.

The Restricted Outcome Space of the Guttman Structure Ω'

Let Ω' be the outcome subspace of all possible Guttman patterns. The number of patterns is $m_i + 1$. In the case $m_i = 3$, the number of response patterns is four; the outcome space Ω' in this case is also shown in Table 6.5.

The next step is distinctive in understanding the implications of the PRM. Suppose there are a large number of essays that have been judged independently at each level. Although it would not be done in practice, for the purpose of the exposition, suppose that only those essays whose judgments conformed to the Guttman pattern were retained. This may seem wasteful of data, but as will be seen when the process is reversed, the implied Guttman structure is the only form of data available.

Moreover, and perhaps remarkably, it will be shown that the threshold estimates obtained from applying the PRM to data of the Guttman subspace are statistically equivalent to those obtained by applying the dichotomous RM to data from the whole space.

Let $X_{ni} = x = \sum_{k=1}^{m_i} y_{nik}$; $x \in \{0, 1, 2, \dots, m_i\}$ be the random variable defined by the sum of the successes across the thresholds for item i . Within a Guttman subspace, in which the thresholds are ordered, the total score recovers each pattern completely. The possible values of this variable are shown in Table 6.6, which shows another outcome space $\Omega'_{0,1}$, to be defined shortly.

An important advantage of taking responses only in the Guttman structure is that a response in a category as in Table 6.3 can be inferred. For example, the response pattern of $x = 0 \equiv (0, 0, 0)$ implies being unsuccessful at P, and at C and at D. Therefore, the response pattern implies the category of F. The response pattern $x = 1 \equiv (1, 0, 0)$ implies being successful at P, but unsuccessful at C and at D. Therefore, this response pattern implies the category of P, and so on. The inferred response category from the Guttman pattern and each total score are shown in Table 6.6. We return to this inference when completing the reverse direction in the derivation, that from the dependent response structure of Table 6.3 to the inferred independent response structure of Table 6.4.

TABLE 6.6 Inferred Response Categories for the Guttman Response Space Ω' and a Subspace for Two Successive Total Scores $\Omega'_{0,1}$

Inferred Category and the Total Score					
$X_{ni} = x$		y_{ni1}	y_{ni2}	y_{ni3}	
0	Fail	0	0	0	$\Omega'_{0,1}$
1	Pass	1	0	0	
2	Credit	1	1	0	Ω'
3	Distinction	1	1	1	

Copyright © 2010, Taylor & Francis Group. All rights reserved.

The sum of the probabilities of the Guttman patterns within the full outcome space Ω is not 1. It is given by

$$D = Q_{ni1}Q_{ni2}Q_{ni3}\dots Q_{nim_i} + P_{ni1}Q_{ni2}Q_{ni3}\dots Q_{nim_i} + P_{ni1}P_{ni2}Q_{ni3}\dots Q_{nim_i} + \dots P_{ni1}P_{ni2}P_{ni3}\dots P_{nim_i} \tag{6.11}$$

Then the probability of each of the Guttman patterns, *conditional on the outcome space Ω'* , is given by simply normalizing the responses within Ω' so that their probabilities do sum to 1, that is,

$$P\{X_{ni} = x|\Omega'\} = P_{ni1}P_{ni2}P_{ni3}\dots P_{nix}Q_{nix+1}Q_{nix+2}\dots Q_{nim_i} / D \tag{6.12}$$

where D is the normalizing factor ensuring that $\sum_{x=0}^{m_i} \Pr\{X_{ni} = x|\Omega'\} = 1$. Although constructed simply as a normalizing factor, because it contains all the terms of the numerator, it is important to the operation of the PRM. It ensures that the probability of a response in any category is a function of the probabilities of all of the other categories. Thus, through the constraint of the Guttman structure, we have a probability structure compatible with that of Table 6.3 in which the response in any category cannot be independent of the response of any other category, there being only one response in one of the categories.

This line of reasoning, beginning with an independent outcome space at thresholds, and then restricting the outcome space to the Guttman structure, is used (Andrich, 1978) in resolving the coefficients of the PRM presented in Rasch (1961) and Andersen (1977), and is applied to a case to demonstrate an interpretation of reversed thresholds in the context of contingency tables (Andrich, 1979). In this chapter we take the understanding of that interpretation a stage further.

In anticipation of this stage, we summarize the development so far: We have (1) reduced an outcome space of m_i experimentally independent dichotomous judgments to a subspace that has the Guttman structure, (2) observed that the responses in this space parallel the responses in one of just $m_i + 1$ ordered categories, as in the general format of Table 6.1 and the specific format of Table 6.3, and (3) recognized that the probability of a response in any category is a function of the probability of the response in all other categories.

The Doubly Conditioned Outcome Space $\Omega'_{x-1,x}$

Consider now an even smaller outcome space, $\Omega'_{x,x-1}$, a subspace of the Guttman space Ω' . It is the response space corresponding to two successive total scores within the Guttman subspace. This outcome space is shown in Table 6.6 for $x = 1$.

We take the ratio of the probabilities of the outcomes within this doubly condition outcome space $\Omega'_{x-1,x}$:

$$\frac{\Pr\{X_{ni} = x|\Omega'\}}{\Pr\{X_{ni} = x-1|\Omega'\}} = \frac{P_{ni1}P_{ni2}P_{ni3}\dots P_{nix}Q_{nix+1}Q_{nix+2}\dots Q_{nim_i}/D}{P_{ni1}P_{ni2}P_{ni3}\dots P_{nix-1}Q_{nix}Q_{nix+1}Q_{nix+2}\dots Q_{nim_i}/D} = \frac{P_{nix}}{Q_{nix}} \quad (6.13)$$

This ratio is simply the odds of a response x rather than $x-1$, giving the probability of a response in category x within the space $\Omega'_{x-1,x}$ of two adjacent categories as

$$\frac{\Pr\{X_{ni} = x|\Omega'\}}{\Pr\{X_{ni} = x-1|\Omega'\} + \Pr\{X_{ni} = x|\Omega'\}} = \frac{P_{nix}}{P_{nix} + Q_{nix}} = P_{nix} \quad (6.14)$$

Notating explicitly the outcome space $\Omega'_{x-1,x}$, Equation 6.14 may be expressed as

$$P_{nix} = \Pr\{X_{ni} = x|\Omega'_{x-1,x}\} = \frac{\Pr\{X_{ni} = x|\Omega'\}}{\Pr\{X_{ni} = x-1|\Omega'\} + \Pr\{X_{ni} = x|\Omega'\}} \quad (6.15)$$

The probability of Equation 6.15 implies a dichotomous response. Clearly, there is no observed experimentally independent dichotomous response within the Guttman subspace. Therefore, this response is *implied* or *latent*. Being latent, however, does not make it any less important. Furthermore, being located in the higher of two ordered categories can be considered the implied successful response relative to the two categories, and the implied response in the lower of the two categories an unsuccessful response. Then the probability of Equation 6.15 is the probability of being successful in adjacent categories within the Guttman subspace Ω' .

The remarkable result in Equation 6.15 is that the *value* P_{nix} in the doubly conditioned sample space $\Omega'_{x-1,x}$, which is the probability of successful responses between the two adjacent categories, is exactly the probability of *success* at the corresponding category in the experimentally independent outcome space Ω ; that is,

$$\Pr\{X_{ni} = x|\Omega'_{x-1,x}\} = \Pr\{Y_{nix} = 1|\Omega\} = P_{nix} \quad (6.16)$$

It is the result of Equation 6.16 that provides unique possibilities for studying the empirical ordering of categories from a response structure that has experimental dependence to one in which experimental independence is implied. To do so unequivocally, we need now to proceed in the reverse direction from that in which we have proceeded so far in this section. That is, since the responses in a format system such as that of Tables 6.1 and 6.3 are not from experimentally independent dichotomous outcomes, but have

only one response in one of only m_i categories, we need to proceed from this response format. However, we are given the cue as to how to proceed from the above derivation: from the experimentally independent response format to a restricted dependent one.

Identification of an Outcome Space From the Definition of Success Between Two Adjacent Categories in a Nonexperimental Design

The notation we use in proceeding in the reverse direction is identical to that used in the above subsections. However, it is stressed that these derivations begin mathematically in their own right.

Thus, let $X_{ni} = x, x \in \{0, 1, 2, \dots, m_i\}$ be the random variable that denotes the response in one of $m_i + 1$ successive categories. This integer variable denotes only order, with successive integers corresponding to successive empirical categories, there being no implication that the sizes of the categories are in any sense the same.

Further, let $\Pr\{X_{ni} = x\}$ be the probability of the response x :

$$\sum_{x=0}^{m_i} \Pr\{X_{ni} = x\} = 1$$

Now define

$$\frac{\Pr\{X_{ni} = x\}}{\Pr\{X_{ni} = x - 1\} + \Pr\{X_{ni} = x\}} = P_{nix} \tag{6.17}$$

and $Q_{nix} = 1 - P_{nix}$.

Clearly the response *implied* by Equation 6.17 is dichotomous relative to the two categories $x - 1$ and x . The response is implied or latent because there is no observed response of the kind—there is only one response in one of the $m_i + 1$ categories. The outcome space of this latent response is denoted $\Omega'_{x-1,x}$ so that Equation 6.17 can be written more explicitly as

$$\Pr\{X_{ni} = x | \Omega'_{x,x-1}\} = P_{nix} \tag{6.18}$$

Because there is only one response in one of the categories, and because x reflects the higher of the two categories, P_{nix} can be considered the conditional probability of an implied *successful response* between categories x and $x - 1$. Then

$$Q_{nix} = 1 - P_{nix} = \Pr\{X_{ni} = x - 1 | \Omega'_{x,x-1}\}$$

is the probability of an *unsuccessful response* between the same two categories.

Wright and Masters (1982) began their derivation of the PRM from Equation 6.17, but did not make explicit the outcome space and introduced the dichotomous Rasch model for P_{nix} prematurely to be able to uncover the full significance of this starting point. It also led them into some errors of interpretation (Andrich, 2005).

To model the observed response in one of the $m_i + 1$ ordered categories in terms of the latent response between a pair of adjacent categories, we need to obtain a statement for $\Pr\{X_{ni} = x\}$. In particular, we need to *infer* an outcome space for $X_{ni} = x, x \in \{0, 1, \dots, m_i\}$. Let this outcome space be Ω' . Making it explicit in Equation 6.17 gives

$$P_{nix} = \frac{\Pr\{X_{ni} = x | \Omega'\}}{\Pr\{X_{ni} = x - 1 | \Omega'\} + \Pr\{X_{ni} = x | \Omega'\}} \quad (6.19)$$

from which $Q_{nix} = 1 - P_{nix}$.

To stress, we begin with Equations 6.17 and 6.18, in which a subspace $\Omega'_{x, x-1}$ is defined that contains two adjacent response categories, and we need to infer a space for the response $X_{ni} = x$, which we denote Ω' .

Let $\pi_x \equiv \Pr\{X_{ni} = x | \Omega'\}$ for convenience in the derivation in which the subscripts n and i are dropped, it being understood that the response is that of a single person to a single item. Similarly, we drop the same subscripts in the probability statements for convenience and let $P_{nix} = P_x$ and $Q_{nix} = Q_x$. Then

$$P_x = \frac{\pi_x}{\pi_{x-1} + \pi_x} \quad (6.20)$$

From Equation 6.20, we need to derive $\pi_x \equiv \Pr\{X_{ni} = x | \Omega'\}$ in terms of P_{nix} and Q_{nix} :

$$P_x (\pi_{x-1} + \pi_x) = \pi_x, \quad \pi_x (1 - P_x) = \pi_{x-1} P_{nix}$$

that is, $\pi_x Q_{nix} = \pi_{x-1} P_{nix}$, giving

$$\pi_x = \pi_{x-1} \frac{P_x}{Q_x} \quad (6.21)$$

Beginning with $\pi_x, x = 1$, the recursive relationship

$$\pi_x = \pi_0 \frac{P_1}{Q_1} \frac{P_2}{Q_2} \frac{P_3}{Q_3} \dots \frac{P_x}{Q_x} = \pi_0 \prod_{k=1}^x \frac{P_k}{Q_k} \quad (6.22)$$

follows. However $\sum_{x=0}^{m_i} \pi_x = 1$; therefore, $\pi_0 = \frac{1}{1 + \sum_{x=1}^{m_i} \prod_{k=1}^x \frac{P_k}{Q_k}}$, and substituting for π_0 in Equation 6.22 gives

$$\pi_x = \frac{\prod_{k=1}^x \frac{P_k}{Q_k}}{1 + \sum_{x=1}^m \prod_{k=1}^x \frac{P_k}{Q_k}} \tag{6.23}$$

That is, in full,

$$\pi_x = \frac{\frac{P_1}{Q_1} \frac{P_2}{Q_2} \frac{P_3}{Q_3} \dots \frac{P_x}{Q_x}}{1 + \frac{P_1}{Q_1} + \frac{P_1}{Q_1} \frac{P_2}{Q_2} + \frac{P_1}{Q_1} \frac{P_2}{Q_2} \frac{P_3}{Q_3} + \dots + \frac{P_1}{Q_1} \frac{P_2}{Q_2} \frac{P_3}{Q_3} \dots \frac{P_m}{Q_m}} \tag{6.24}$$

which on simplification gives $\pi_x = P_1 P_2 P_3 \dots P_x Q_{x+1} Q_{x+2} \dots Q_m / D$, where

$$D = Q_1 Q_2 Q_3 \dots Q_m + P_1 Q_2 Q_3 \dots Q_m + P_1 P_2 Q_3 \dots Q_m + \dots + P_1 P_2 P_3 \dots P_m \tag{6.25}$$

That is, in full

$$P\{X_{ni} = x | \Omega'\} = P_{ni1} P_{ni2} P_{ni3} \dots P_{nix} Q_{nix+1} Q_{nix+2} \dots Q_{nim_i} / D \tag{6.26}$$

Equation 6.26, which gives the probability of a response in any category x , implies a successful or unsuccessful response between *every* pair of successive categories (and not just, for example, the first x categories). Further, it implies a particular structure of successful and unsuccessful responses for each response $X_{ni} = x$. This structure, which should be obvious, is formalized for completeness below.

Let $Y_{nix}, x = 1, 2, \dots, m_i$ be a sequence of dichotomous random variables, $Y_{nix}, y; y \in \{0, 1\}$, which correspond to the successes and failures implied on the right side of Equation 6.26, and let $\Pr\{Y_{nix} = 1\} = P_{nix}; Q_{nix} = 1 - P_{nix}$. At this point we specify nothing about the independence or dependence of these responses.

If $\Pr\{X_{ni} = x | \Omega'\} = P_{ni1} P_{ni2} P_{ni3} \dots P_{nix} Q_{nix+1} Q_{nix+2} \dots Q_{nim_i} / D$, then according to the definition of the random variable $Y_{nix} = y, y \in \{0, 1\}$ above,

$$\begin{aligned} \Pr\{X_{ni} = x | \Omega'\} &= [\Pr\{Y_{ni1} = 1\} \Pr\{Y_{ni2} = 1\} \dots \Pr\{Y_{nix} = 1\} \\ &\quad \times \Pr\{Y_{nix+1} = 0\} \Pr\{Y_{nix+2} = 0\} \dots \Pr\{Y_{nim_i} = 0\} | \Omega'] / D \end{aligned}$$

Thus, $X_{ni} = x$, which arises with the pattern of probabilities of success and failure in Equation 6.26, is consistent with the vector of responses

$$(Y_{ni1}, Y_{ni2}, Y_{ni3}, \dots, Y_{nix}, Y_{nix+1}, Y_{nix+2}, \dots, Y_{nim_i}) = \underbrace{\{1, 1, 1, \dots, 1\}}_x, \underbrace{\{0, 0, 0, \dots, 0\}}_{m_i - x}$$

This implies that the outcome space Ω' , which was to be inferred from Equation 6.19, is the Guttman structure. Only $m_i + 1$ such patterns are possible, which is exactly the number of response categories. It is stressed that this Guttman space Ω' is *latent*, not a manifest space. There is no observed response between any adjacent categories. It is also inferred as the outcome space starting with Equation 6.19.

Inferring an Experimentally Independent Outcome Space Ω

Given the Guttman space Ω' , we infer the existence of a complete space Ω of which Ω' is a subspace. In this complete space we can infer experimentally independent responses. Thus, from the construction of Equation 6.26, it is clear that

$$\sum_{x=0}^{m_i} \Pr\{X_{ni} = x | \Omega'\} = \sum_{x=0}^{m_i} P_{ni1} P_{ni2} P_{ni3} \dots P_{nix} Q_{nix+1} Q_{nix+2} \dots Q_{nim_i} / D = 1 \quad (6.27)$$

and therefore that

$$\sum_{x=0}^{m_i} P_{ni1} P_{ni2} P_{ni3} \dots P_{nix} Q_{nix+1} Q_{nix+2} \dots Q_{nim_i} = D \neq 1 \quad (6.28)$$

Now consider all 2^{m_i} patterns of responses $(Y_{ni1}, Y_{ni2}, Y_{ni3}, \dots, Y_{nix}, Y_{nix+1} \times Y_{nix+2}, \dots, Y_{nim_i})$ in which $Y_{nix} = y \in \{0, 1\}$ of which the $m_i + 1$ Guttman patterns $\left(\underbrace{1, 1, \dots, 1}_x, \underbrace{0, 0, \dots, 0}_{m_i - x} \right)$ are a subspace. Let this outcome space be denoted Ω . Then by the definition of Y_{nix} and Equation 6.28, it can be shown readily that the probabilities of the set of all possible 2^{m_i} patterns $\{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i}) | \Omega\}$ sum to 1:

$$\sum_{\Omega} \Pr\{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})\} = \sum_{\Omega} \prod_{x=1}^{m_i} P_{nix}^{y_{nix}} Q_{nix}^{1-y_{nix}} = 1 \quad (6.29)$$

However, Equation 6.29 implies experimental independence of responses in the sense that

$$\Pr\{(y_{ni1}, y_{ni2}, \dots, y_{nix}, \dots, y_{nim_i})\} = P(y_{ni1}) P(y_{ni2}) \dots P(y_{nim_i}) \quad (6.30)$$

We reason that given that there is a Guttman outcome space Ω' with the $m_i + 1$ response patterns, and with the probability structure of Equation 6.26, it must be the *subspace* of a hypothesized, experimentally independent outcome space Ω with 2^{m_i} response patterns.

In summary, the *values* of the probabilities of successes $P_{nix}, x = 1, 2, 3, \dots, m_i$, which hold in the outcome space $\Omega'_{x-1, x}$ for two successive categories within the outcome space Ω' , hold in the inferred experimentally independent and

complete space Ω . This means that if the thresholds can be estimated from the Guttman sample space Ω' according to Equation 6.26, then the values of the thresholds can be inferred to be statistically equivalent to those in an experimentally independent outcome space Ω compatible with Ω' .

In this inferred experimentally independent outcome space, as we argued in the previous section, we require the natural ordering of threshold difficulties to hold.

It is stressed that if one set of data were collected experimentally according to the design of Table 6.4, and another set were collected according to the ordered response categories in the usual design of Table 6.3 for the same essays and the same definition of the categories, nothing in the above analysis would guarantee that the thresholds would be the same from the two data collections. Whether or not they give equivalent results in any particular assessment is an empirical question. It might be interesting to conduct such experiments. However, the analysis implies that there is a hypothetical experimentally independent outcome space Ω of dichotomous random variables with inferred probabilities of which the Guttman outcome space Ω' is a subspace. In any complete space Ω , the ordering of the threshold difficulties needs to be in the natural order of the categories if the categories are working empirically as intended.

Construction and Interpretation of the PRM

The above analysis of outcome spaces has not rested on the dichotomous RM. However, our earlier specification (discussed above) was that the latent thresholds should have the same relative difficulties irrespective of the location of essays on the continuum, and with the higher-ordered category reflecting a greater location of the essays on the continuum. This, we concluded, implied the responses should conform to the dichotomous RM. Now we bring the dichotomous RM and the analysis of outcome spaces in the previous section together.

Because the latent Guttman structure arises as the outcome space irrespective of the starting point of the derivation, we begin with Equation 6.26:

$$\Pr\{X_{ni} = x|\Omega'\} = P_{ni1}P_{ni2}P_{ni3}\dots P_{nix}Q_{nix+1}Q_{nix+2}\dots Q_{nim_i}/D \tag{6.31}$$

Let

$$P_{nix} = \frac{e^{\beta_n - \delta_{ix}}}{1 + e^{\beta_n - \delta_{ix}}} \tag{6.32}$$

and

$$Q_{nix} = 1 - \frac{e^{\beta_n - \delta_{ix}}}{1 + e^{\beta_n - \delta_{ix}}} = \frac{1}{1 + e^{\beta_n - \delta_{ix}}}$$

which is the dichotomous RM with the threshold δ_{ix} made explicit.

Inserting Equations 6.32 and 6.31 gives

$$\Pr\{X_{ni} = x|\Omega'\} = \frac{e^{\beta_n - \delta_{i1}}}{1 + e^{\beta_n - \delta_{i1}}} \frac{e^{\beta_n - \delta_{i2}}}{1 + e^{\beta_n - \delta_{i2}}} \cdots \frac{e^{\beta_n - \delta_{ix}}}{1 + e^{\beta_n - \delta_{ix}}} \frac{1}{1 + e^{\beta_n - \delta_{ix+1}}} \cdots \frac{1}{1 + e^{\beta_n - \delta_{im_i}}} / D$$

which, on making D explicit, simplifies to

$$\Pr\{X_{ni} = x|\Omega'\} = e^{x\beta_n - \sum_{k=0}^x \delta_{ik}} / \gamma_{ni} \quad (6.33)$$

where $\delta_{k0} \equiv 0$ is used for notational convenience, and

$$\gamma_{ni} = \sum_{x=0}^{m_i} e^{x\beta_n - \sum_{k=0}^x \delta_{ik}} \quad (6.34)$$

is the normalizing factor in Equation 6.33 ensuring that the sum of its probabilities is 1.

Equation 6.33 is a general form of the PRM.

Other Parameterizations

Because the thresholds are identified with each item, it is often convenient to consider the thresholds of an item as deviations from its overall location. Let $\delta_{ik} = \delta_i + \tau_{ik}$, where $\sum_{k=0}^{m_i} \tau_{ik} = 0$. Then δ_i is the mean of the thresholds δ_{ik} , and τ_{ik} are thresholds that are deviations from δ_i . This gives Equation 6.33 in the form

$$\Pr\{X_{ni} = x|\Omega'\} = e^{x\beta_n - x\delta_i - \sum_{k=0}^x \tau_{ik}} / \gamma_{ni} = e^{x(\beta_n - \delta_i) - \sum_{k=0}^x \tau_{ik}} / \gamma_{ni} \quad (6.35)$$

This is also a convenient form of the model for estimation of the threshold parameters that can be reparameterized into principal components (Andrich & Luo, 2003), of which δ_i is the first principal component. Further, this reparameterization permits estimates of thresholds to be obtained even when some categories have zero frequencies.

Because the total score $r_n = \sum_{i=1}^I x_{ni}$ is the sufficient statistic for the person parameter β_n (Andersen, 1977; Andrich, 1978), this estimation can be carried out conditionally, eliminating the person parameters. As indicated earlier, we do not deal in this chapter with estimation and tests of fit, and concentrate on the hypothesis of the empirical ordering of the categories as manifested in the threshold estimates, and the understanding of the operation of the model

in relation to these threshold estimates. However, we note that the thresholds are estimated independently of the person parameters, and therefore independently of any distribution of these parameters. This means that the threshold estimates *reflect structural relationships among each other* and not the distribution of the person parameters.

Equation 6.35 is also convenient if it is hypothesized that the distances between thresholds might be the same across items, with the only difference among the item parameters being the location of the item determined by the mean of the thresholds. The thresholds are then not subscripted by the item parameter, giving

$$\Pr\{X_{ni} = x|\Omega'\} = e^{x(\beta_n - \delta_i) - \sum_{k=0}^x \tau_k} / \gamma_{ni} \quad (6.36)$$

It is this parameterization that is sometimes referred to as the *rating scale model*, while it is the parameterization of Equation 6.33, which is sometimes referred to as the *partial credit model*. It should be clear that the response model is the same in Equations 6.36 and 6.33, and that the only difference is in the parameterization. Whether the thresholds are equidistant among items is again an empirical question, not a response process question.

An important observation from Equation 6.33, which reflects the dependence among the responses, is that the probability of a response in any category $\{X_{ni} = x|\Omega'\}$ is a function of all the thresholds, and not just of an adjacent pair defining a category. This is easily seen from the denominator $\gamma_{ni} = \sum_{x=0}^{m_i} e^{x\beta_n - \sum_{k=0}^x \delta_{ik}}$ in Equation 6.33. The denominator contains all thresholds, so a change of the value of any one threshold will change the probability of a response in every category. In particular, the probability of a response in the first category is a function of the last threshold. This confirms the structural dependence in the responses among categories.

Identity of the Dichotomous RM in the Full Ω Space and the Guttman Space Ω'

For completeness, we now summarize the relationships among the parameters of the PRM according to Equation 6.16. Inserting the dichotomous RM for P_{nix} , we have

$$\Pr\{X_{ni} = x|\Omega'_{x-1,x}\} = \Pr\{Y_{nix} = 1|\Omega\} = \frac{e^{\beta_n - \delta_{ix}}}{1 + e^{\beta_n - \delta_{ix}}} \quad (6.37)$$

Thus, using the example of grades awarded in Figure 6.1 and Table 6.4, the explicit relationships between the responses in the outcome spaces and the parameters are shown in Table 6.7. In particular, the last row of Table 6.7 shows that the threshold δ_{ix} , at which the probability of a successful response is 0.5 at category $x, x > 0$ in the experimentally independent outcome space

TABLE 6.7 Equivalences of Corresponding Thresholds in the Spaces Ω and $\Omega'_{x-1,x}$

Outcome Space	Response	$\mathbf{x} = \mathbf{P}$	$\mathbf{x} = \mathbf{C}$	$\mathbf{x} = \mathbf{D}$
$\Pr\{Y_{nix} = 1 \mid \Omega\}$		$\frac{e^{\beta_n - \delta_{iP}}}{1 + e^{\beta_n - \delta_{iP}}}$	$\frac{e^{\beta_n - \delta_{iC}}}{1 + e^{\beta_n - \delta_{iC}}}$	$\frac{e^{\beta_n - \delta_{iD}}}{1 + e^{\beta_n - \delta_{iD}}}$
$\Pr\{X_{ni} = x \mid \Omega'_{x-1,x}\}$		$\frac{e^{\beta_n - \delta_{iFP}}}{1 + e^{\beta_n - \delta_{iFP}}}$	$\frac{e^{\beta_n - \delta_{iPC}}}{1 + e^{\beta_n - \delta_{iPC}}}$	$\frac{e^{\beta_n - \delta_{iCD}}}{1 + e^{\beta_n - \delta_{iCD}}}$
		$\delta_{iP} = \delta_{iFP}$	$\delta_{iC} = \delta_{iPC}$	$\delta_{iD} = \delta_{iCD}$
$\{\delta_{ix} \mid \Omega\} = \{\delta_{ix-1,x} \mid \Omega'\}$				

Ω , is identical to the threshold $\delta_{ix-1,x}$, at which the probability of a successful response $x > 0$ relative to the adjacent category $x - 1$ is 0.5 in the constrained outcome space $\Omega'_{x-1,x}$.

Consolidating the Interpretation With a Simulation Study

To consolidate the derivations, we now show analyses of the same data in the different response spaces according to the dichotomous RM and the PRM. Table 6.8 summarizes a data set in which 5,000 performances from a normal distribution, $\beta \approx N(0,1)$, were simulated as having been assessed in the *experimentally independent* design of Table 6.4. The responses followed the dichotomous RM *independently* at each threshold. There were three thresholds in this design corresponding to four categories in the standard design, as in Table 6.3. Two sets of three independent assessments were simulated, conceptualized as responses from three expert and three novice judges, respectively. In the former, the successive thresholds have increasing values as required. For the novices, however, the values of the first and second thresholds are reversed, simulating an effect where Novice Judges 1 and 2 have confused their relative standards. These generating parameters are shown in Table 6.8. Table 6.8 also shows two sets of estimates. The first was obtained from the dichotomous RM applied to data in the full space Ω . These are pairwise conditional estimates in Andrich and Luo (2003). They are consistent

TABLE 6.8 Generating Parameters and Estimates of Independent Dichotomous Items Using the Dichotomous RM on the Full Space and the PRM in the Guttman Subspace

Judges	Item Number	Generating Locations	Estimates from the Dichotomous RM in a Full Space	Estimates from the PRM in a Guttman Space
Novice 1 (P)	1	-1.0	-1.001	-1.034
Novice 2 (C)	2	0.0	0.031	0.114
Novice 3 (D)	3	1.0	0.941	0.920
Expert 1 (P)	4	0.0	0.009	-0.024
Expert 2 (C)	5	-1.0	-1.018	-1.000
Expert 3 (D)	6	1.0	1.037	1.024

(Zwinderman, 1995). The second was obtained from the *subset* of data in the space Ω' whose responses conformed to a Guttman structure for both the experts and the novices. The Guttman structure was defined *according to the intended, hypothetical ordering of the thresholds*. Thus for experts, it was based on correct threshold ordering; for the novices, it was based on incorrect ordering. From each Guttman pattern, a single score was formed according to $X_{ni} = \sum_{k=1}^3 Y_{nik}$, giving one assessment for the experts and novices, each with a maximum score of 3. The estimates are from the PRM applied to this subset of data. Without elaborating the statistical properties of the estimates, it is evident that the thresholds are recovered excellently in both data sets with the respective models. They are not the same in part because the PRM data are a subset of the dichotomous RM data—all responses that did not conform to the hypothesized Guttman pattern were excluded. That the former are recovered excellently is as expected. However, it is reassuring of the theory developed in this chapter that the subset of data that conforms to the intended Guttman pattern also recovers the estimates excellently, including those with reversed thresholds.

Of course, in real data from a typical design, only the analysis from the PRM is available. However, as demonstrated above, the threshold estimates can be interpreted as if they had arisen from a design, compatible with the data analyzed by the PRM, in which independent judgments were made at each threshold as in the design of Table 6.4. And in this design, it is required that the thresholds are in the natural order.

Figure 6.3 shows the estimated category characteristic curves (CCCs) from the PRM for the experts and the novices. For the experts, where the thresholds are in the correct order, it is evident that all categories have a region in which they have higher probability of occurring than any other category. For the novices, where the thresholds are not in the correct order, the category with a score of 1 (pass) never has a higher probability of occurring than the scores of 0 (fail) and 2 (credit). Even where it has the highest probability of occurring, near -0.5 logits, *simultaneously* the scores of 0 (fail) and 2 (credit) have a higher probability of occurring. This is a symptom of a problem with the empirical operation of the categories, a symptom that in this simulated data set we know is present. The argument in this chapter is that such CCCs in real data also imply a problem in the ordering of the categories.

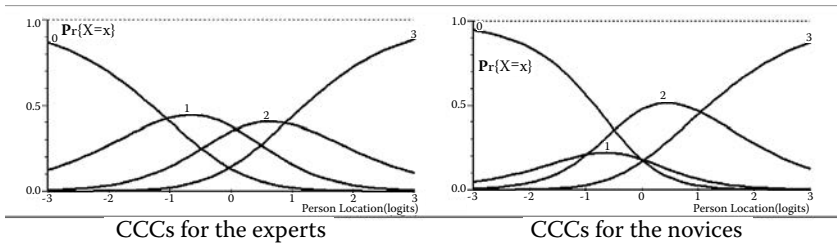


FIGURE 6.3 Category characteristic curves (CCCs) for the experts and novices in simulated data.

Derivation of the PRM Using the 2PLM

In order to better understand the PRM and its properties, it is now derived beginning with the 2PLM of Equation 6.4 rather than the dichotomous RM in Equation 6.2. This is the construction presented in detail in Andrich (1978). It also helps make more explicit its incompatibility with the GRM.

Inserting the 2PLM of Equation 6.4 into Equation 6.26 gives

$$\begin{aligned} \Pr\{X_{ni} = x|\Omega'\} &= \frac{e^{\alpha_{i1}(\beta_n - \delta_{i1})}}{1 + e^{\alpha_{i1}(\beta_n - \delta_{i1})}} \frac{e^{\alpha_{i2}(\beta_n - \delta_{i2})}}{1 + e^{\alpha_{i2}(\beta_n - \delta_{i2})}} \cdots \frac{e^{\alpha_{ix}(\beta_n - \delta_{ix})}}{1 + e^{\alpha_{ix}(\beta_n - \delta_{ix})}} \\ &\times \frac{1}{1 + e^{\alpha_{ix+1}(\beta_n - \delta_{ix+1})}} \cdots \frac{1}{1 + e^{\alpha_{im_i}(\beta_n - \delta_{im_i})}} / D \end{aligned} \tag{6.38}$$

On simplification of the numerator and summarizing,

$$\begin{aligned} \Pr\{X_{ni} = x|\Omega'\} &= \frac{e^{(\alpha_{i1} + \alpha_{i2} + \dots + \alpha_{ix})\beta_n - \sum_{k=1}^x \alpha_{ix} \delta_{ix} \alpha_{i1} \delta_{i1} - \alpha_{i2} \delta_{i2} - \dots - \alpha_{ix} \delta_{ix}}}{\prod_{k=1}^{m_i} (1 + e^{\alpha_{ix}(\beta_n - \delta_{ix})})} / D \\ &= \frac{e^{\varphi_{ix} \beta_n + \kappa_{ix}}}{\prod_{k=1}^{m_i} (1 + e^{\alpha_{ix}(\beta_n - \delta_{ix})})} / D \end{aligned} \tag{6.39}$$

where

$$\varphi_{ix} = \alpha_{i1} + \alpha_{i2} + \dots + \alpha_{ix} \tag{6.40}$$

and

$$\kappa_{ix} = -\alpha_{i1} \delta_{i1} - \alpha_{i2} \delta_{i2} - \dots - \alpha_{ix} \delta_{ix} \tag{6.41}$$

The term φ_{ix} was called the scoring function of category x by Rasch (1961), who derived the model generically from a vector form of the model. Andersen (1977) showed that for sufficiency to hold in a unidimensional model, these coefficients had to have the constraint

$$\varphi_{ix} - \varphi_{ix-1} = \varphi_{ix+1} - \varphi_{ix} \tag{6.42}$$

Relationship to Other Models: The scoring function is a very useful concept for defining polytomous Rasch models. Building on Rasch’s (1961) definition of the function as category scores that needed to be estimated, Andrich (1978) clarified the meaning of the scoring function as being the sum of threshold discriminations. When these discriminations are fixed at 1.0, their value is equivalent to integer scored ordered response categories—following Rasch’s original meaning. While the scoring function formulation of Rasch models was originally applied by Andrich to the rating scale model, Muraki (1992) used this formulation to develop the generalized partial credit model, and Adnrich (1988) also showed how the PCM itself can be expressed in a scoring function formulation. This formulation is elaborated below.

It is evident that if the discriminations at the thresholds are made equal, and defined to be 1 arbitrarily and conveniently, that is, $\alpha_{i1} = \alpha_{i2} = \dots = \alpha_{ix} = 1$,

then $\varphi_{ix} = x_i$ is the integer scoring function satisfying Equation 6.42, and further that $\kappa_{ix} = -\delta_{i1} - \delta_{i2} \dots - \delta_{ix}$, giving the PRM of Equation 6.33. Thus, the integer scoring function $\varphi_{ix} = x_i$ of the successive categories arises from the *identity of discrimination* at the thresholds and not from the equality of distances between thresholds, which of course are estimated. This integer scoring gives sufficiency, with the dichotomous RM a special case.

However, the construction beginning with the 2PLM is instructive regarding combining categories. Andersen (1977) also clarified Rasch's (1966) observation that the probabilities of responses in adjacent categories could not be summed while retaining the structure of the model of Equation 6.39 by showing that the probabilities of two adjacent categories $x-1$ and x could only be combined if $\varphi_{ix-1} = \varphi_{ix}$. This abstract conclusion can be understood from Equation 6.40 readily, but distinctively, by noting that if

$$\varphi_{ix-1} = \alpha_{i1} + \alpha_{i2} + \dots \alpha_{ix-1} = \varphi_{ix} = \alpha_{i1} + \alpha_{i2} + \dots \alpha_{ix-1} + \alpha_{ix} \tag{6.43}$$

then $\alpha_{ix} = 0$. That is, the probabilities of two adjacent categories can be summed *only if the discrimination between the two categories is 0*, that is, if the responses between the two categories are random for all values of β . This seems eminently sensible. This property of the model has been discussed in detail in Jansen and Roskam (1986) and Andrich (1995).

In particular, pooling frequencies of adjacent categories so that there are just two new categories, that is, *dichotomizing the categories*, is not compatible with the PRM. Thus, there is a distinction between *dichotomizing the continuum*, as is done in constructing the experimental design in Table 6.4 and dichotomizing the categories.

Of course, when data do not fit the model or when thresholds are reversed, exploratory analysis that might involve pooling categories in such a way may be instructive in understanding the mis-fit or the reversed thresholds.

Furthermore, it is evident from Equation 6.41 that if the discriminations at the thresholds are *different in the data* so that the responses are effectively governed by products of the thresholds and discriminations at the thresholds, that is, $\kappa_{ix} = -\alpha_{i1}\delta_{i1} - \alpha_{i2}\delta_{i2} - \dots - \alpha_{ix}\delta_{ix}$, and if the data are analyzed according to the PRM in which the discriminations are identical, that is, $\kappa_{ix} = -\delta_{i1} - \delta_{i2} - \dots - \delta_{ix}$, then the threshold parameter estimates will be affected by the different discriminations in the data. In particular, this effect on the threshold locations will manifest itself when the discrimination between a pair of adjacent categories is 0—when the responses between two adjacent categories are random. In some cases where such disturbance of the model is present, reversed threshold estimates can appear.

Conclusion

The chapter studies the PRM from first principles demonstrating the implications of its two distinctive properties: first, that the thresholds that partition the continuum into contiguous categories when the categories are working

as intended can be reversed in data and then not define a category; second, that the category frequencies and probabilities of adjacent categories cannot be summed routinely to form a new category arbitrarily.

Regarding the second property, it is shown that two adjacent categories can be combined in the above way only if the discrimination between the adjacent categories is zero, that is, if the response between the two categories is random. In this way, the model is shown to be different from the alternate model for ordered categories, the GRM, which is constructed by combining categories in just this way.

Most of the space in the chapter was devoted to the first of the properties, the possible reversal of threshold estimates. First, it was argued that in the case where a design could be constructed in which the responses at the thresholds were independent, it would be required that the threshold difficulties show the intended order of the successive categories. Second, it was then shown that the reversal of threshold estimates could be interpreted in a response outcome space in which responses at the thresholds were independent. Therefore, third, it was concluded that the reversed threshold estimates indicated a problem with the empirical ordering of the categories.

This argument was set in the context of a theme of the chapter, given a direction from Fisher, that the ordering of the categories should be a property of the data, and not a property of the model by which they are analyzed. Thus, the empirical ordering of the thresholds becomes an empirical question, not simply an assertion. The property of possible reversals of the threshold estimates in the PRM was again contrasted to the corresponding property of the GRM in which the ordering of its thresholds, defined differently from the way they are defined in the PRM, is a property of the model irrespective of the properties of the data.

In the process of the development of this argument, three possible functions of statistical models were summarized: (1) accounting for and describing data evidenced by tests of fit, (2) characterizing a process, and (3) characterizing a requirement. In the case of the PRM, it was stressed that it was not derived from an attempt to model data, but from an attempt to characterize the requirement of invariant comparisons among persons and among items. Thus, the case for the model does not rest on any test of fit, and whether the data fit the model or not is an empirical question. One of the points that the chapter raises concerns the lack of any previously specified criterion other than that implied in Fisher's approach for the ordering of categories.

It is suggested here that the answer can be found in the dominant paradigm in statistical analysis—that the case for a model is that it accounts for the data at hand. If the model does not account for the data at hand, another model is sought (Andrich, 2004). It is so dominant that it is even used in the case when the model has been derived to characterize a process or when it is derived to meet requirements, such as the PRM. That is, it is seen that the case of accounting for data is both necessary and sufficient for it to be utilized. There seems no other explanation as to why the diagrams such as those for the novices in Figure 6.3, which invite questioning the operation of

the categories, have not been questioned outside the framework of the PRM, and often ignored or circumvented even within that framework. That is, it seems that the reason they have not been questioned is that tests of fit can be found in which the data seem to be accounted for by the model. It should be argued as forcefully as possible that in the case where a model is derived from criteria other than for the purpose of modeling a particular data set, for example, to meet a requirement and to characterize a process such as the PRM, statistical tests of fit are not sufficient.

One final comment is made regarding the person estimates. It might be suggested that in view of the model being able to estimate reversed thresholds that can be interpreted in an experimentally independent design, the model should therefore simply be used to estimate person locations without concern for the ordering of the thresholds. There are two responses to this superficial suggestion. First, the reversal of the thresholds tells that there is something wrong with the intended ordering of the categories, and that signals that there is a problem with the data. The reversed thresholds are a symptom of a problem in the data that the model exposes. The person estimates contain whatever problem is in the data that leads to such estimates.

The second is that the reversed threshold estimates give an indication where the attempt at understanding what it means to have more of the property has failed. It gives an opportunity to better understand the variable and to construct more rigorous items. To use the model simply to account for the reversed thresholds in the data would again be taking the position that the fit of the data to the model is both a necessary and sufficient case for the application of the model.

Acknowledgments

This chapter has benefited from many discussions over the years with many people on the implications of the reversals of Rasch thresholds in polytomous items, and in particular with Guanzhong Luo. The research reported in this chapter was supported in part by Australian Research Council Linkage grants with the Performance Measurement and Reporting Task Force of the Australian National Ministerial Council on Employment, Education, Training and Youth Affairs; UNESCO's International Institute for Educational Planning; the Australian Council for Educational Research; and Pearson Research and Assessment as Industry Partners.

References

- Andersen, E. B. (1973). Conditional inference for multiple choice questionnaires. *British Journal of Mathematical and Statistical Psychology*, 26, 31–44.
- Andersen, E. B. (1977). Sufficient statistics and latent trait models. *Psychometrika*, 42, 69–81.
- Andrich, D. (1978). A rating formulation for ordered response categories. *Psychometrika*, 43, 561–574.

- Andrich, D. (1979). A model for contingency tables having an ordered response classification. *Biometrics*, 35, 403–415.
- Andrich, D. (1985). An elaboration of Guttman scaling with Rasch models for measurement. In N. Brandon-Tuma (Ed.), *Sociological methodology* (Chap. 2, pp. 33–80). San Francisco: Jossey-Bass.
- Andrich, D. (1995). Models for measurement, precision and the non-dichotomization of graded responses. *Psychometrika*, 60, 7–26.
- Andrich, D. (2002). Understanding resistance to the data-model relationship in Rasch's paradigm: A reflection for the next generation. *Journal of Applied Measurement*, 3, 325–357.
- Andrich, D. (2004). Controversy and the Rasch model: A characteristic of incompatible paradigms? *Medical Care*, 42, 7–16.
- Andrich, D. (2005). The Rasch model explained. In S. Alagumalai, D. D. Curtis, & N. Hungi (Eds.), *Applied Rasch measurement: A book of exemplars* (pp. 308–328). Norwell, MA: Springer-Kluwer.
- Andrich, D., & Luo, G. (2003). Conditional estimation in the Rasch model for ordered response categories using principal components. *Journal of Applied Measurement*, 4, 205–221.
- Birnbaum, A. (1968). Some latent trait models and their use in inferring an examinee's ability. In F. M. Lord & M. R. Novick, *Statistical theories of mental test scores* (pp. 397–545). Reading, MA: Addison-Wesley.
- Bock, R. D. (1975). *Multivariate statistical methods in behavioral research*. New York: McGraw-Hill.
- Bock, R. D., & Jones, L. V. (1968). *The measurement and prediction of judgement and choice*. San Francisco: Holden Day.
- Fischer, G. H., & Molenaar, I. W. (Eds.). (1995). *Rasch models: Foundations, recent developments, and applications*. New York: Springer.
- Fisher, R. A. (1958). *Statistical methods for research workers* (13th ed.). New York: Hafner.
- Guttman, L. (1950). The basis for scalogram analysis. In S. A. Stouffer, L. Guttman, E. A. Suchman, P. F. Lazarsfeld, S. A. Star, & J. A. Clausen (Eds.), *Measurement and prediction* (pp. 60–90). New York: Wiley.
- Harris, J. (1991). *Consequences for social measurement of collapsing adjacent categories with three or more ordered categories*. Master of Education Dissertation, Murdoch University, Western Australia.
- Jansen P. G. W., & Roskam, E. E. (1986). Latent trait models and dichotomization of graded responses. *Psychometrika*, 51, 69–91.
- McCullagh, P. (1980). Regression models for ordinal data. *Journal of the Royal Statistical Society, Series B*, 42, 109–142.
- Mellenbergh, G. J. (1995). Conceptual notes on models for discrete polytomous item responses. *Applied Psychological Measurement*, 19, 91–100.
- Rasch, G. (1960). *Probabilistic models for some intelligence and attainment tests*. (Copenhagen, Danish Institute for Educational Research). Expanded edition (1980) with foreword and afterword by B.D. Wright, (1980). Chicago: The University of Chicago Press. Reprinted (1993) Chicago: MESA Press
- Rasch, G. (1961). On general laws and the meaning of measurement in psychology. In J. Neyman (Ed.), *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability IV* (pp. 321–334). Berkeley: University of California Press.
- Rasch, G. (1966). An individualistic approach to item analysis. In P. F. Lazarsfeld & N. W. Henry (Eds.), *Readings in mathematical social science* (pp. 89–108). Chicago: Science Research Associates.

- Samejima, F. (1969). Estimation of latent ability using a response pattern of graded scores. *Psychometric Monographs*, 34 (2, No. 17).
- Samejima, F. (1996). Graded response model. In W. J. van der Linden & R. K. Hambleton (Eds.), *Handbook of item response theory* (pp. 85–100). New York: Springer.
- Samejima, F. (1997). Departure from normal assumptions: A promise for future psychometrics with substantive mathematical modelling. *Psychometrika*, 62, 471–493.
- Thissen, D., & Steinberg, L. (1986). A taxonomy of item response models. *Psychometrika*, 51, 567–577.
- Thurstone, L. L. (1925). A method of scaling psychological and educational tests. *Journal of Educational Psychology*, 16, 433–451.
- Thurstone, L. L. (1928). Attitudes can be measured. *American Journal of Sociology*, 33, 529–554.
- Thurstone, L. L., & Chave, E. J. (1929). *The measurement of attitude*. Chicago: University of Chicago Press.
- Wright, B. D. (1980). Foreword. In G. Rasch, *Probabilistic models for some intelligence and attainment tests*. (Copenhagen, Danish Institute for Educational Research, 1960). Expanded edition (1980), Chicago: The University of Chicago Press. Reprinted (1993) Chicago: MESA Press.
- Wright, B. D. (1997). A history of social science measurement. *Educational Measurement: Issues and Practice*, 16, 33–45.
- Wright, B. D., & Masters, G. N. (1982). *Rating scale analysis: Rasch measurement*. Chicago: MESA Press.
- Wright, B. D., & Panchapakesan, N. (1969). A procedure for sample-free item analysis. *Educational and Psychological Measurement*, 29, 23–48.
- Zwinderman, A. H. (1995). Pairwise parameter estimation in Rasch models. *Applied Psychological Measurement*, 19, (4) 369–375.